



APPLIED MATHS (241)

MOST IMPORTANT & FREQUENTLY TESTED TOPICS

PREPARED BY THUNDERSTUDY

Based on extensive analysis of CBSE question papers and marking schemes (2022-2025).

1. 💰 Financial Mathematics

- **EMI Calculations:** Mastery of Flat Rate vs. Reducing Balance methods is mandatory for 3-5 mark questions.
- **Depreciation:** Focus on Linear Method to find Annual Depreciation, Scrap Value, and Book Value.
- **Growth & Perpetuity:** CAGR (Compound Annual Growth Rate) and the Present Value of Perpetuity.
- **Sinking Funds:** Periodic payments required for future asset replacement.

2. 📈 Calculus & its Applications

- **AOD (Applications of Derivatives):** Strictly Increasing/Decreasing intervals and Local Maxima/Minima.
- **Economic Functions:** Marginal Cost (MC), Average Cost (AC), and Revenue relationships.
- **Differential Equations:** Solving first-order equations and matching order/degree.
- **Integration:** Definite integrals and Area Bounded by curves (application-based).

3. 🎲 Probability & Inferential Statistics

- **Probability Distributions:** Poisson, Normal (Standard Z-scores), and Binomial distributions.
- **Inferential Statistics:** Student's t-test, rejection/acceptance of Null Hypothesis (H_0), and Degrees of Freedom.
- **Expectation:** Calculating Mean and Variance of random variables.

4. | | | |---|---| | 1 | 2 | | 3 | 4 | Algebra (Matrices & Determinants)

- **Linear Systems:** Solving equations using **Cramer's Rule** or **Matrix Method** ($A^{-1}B$).
- **Matrix Properties:** Inverses, Symmetric, and Skew-Symmetric matrices.

5. LPP (Linear Programming)

- **Graphical Method:** Formulating constraints, shading feasible regions, and corner-point optimization.

6. Time Series Analysis

- **Method of Least Squares:** Fitting straight lines ($Y = a + bX$) and forecasting future trends.
- **Moving Averages:** calculating 3-yearly, 4-yearly, or 5-yearly trends accurately.

7. Numerical Applications

- **Quantification:** Clock problems, Boats & Streams, Pipes & Cisterns, and Linear Races.
- **Modular Arithmetic:** Basic congruences and properties.

SCORE 100/100 WITH THUNDERSTUDY